

Roll no. :

24B0973

Name. :

Krishna Sai Shishir

Seat. :

LA001-YELLOW-M29

Marks obtained



CS213/293 Data Structure and Algorithms 2025

IITB India

Midsem 1

Ashutosh Gupta

Compile date: 2025-09-19

Duration : 120 minutes

Total marks : 8 + 4 + 4 + 6 + 8 + 8 + 12 = 50 points

Section A

1. (8 marks) Mark the following statements True / False. Also provide justification.

(a) The amortized worst-case running time of n operations on UnionFind with path compression is $O(n^2)$.

(b) HEAPIFY always pushes down the violating key to the leaf.

(c) At most height change of the root of a tree by rotation on some node in a binary search tree is 1.



(d) The successor of a node in BST always have a distinct key.

(e) The preorder walk takes $O(\log n)$ time on a binary tree with n nodes.

(f) Linear probing is a special case of open addressing.

(g) $O(n) \subset O(n^2)$

(h) The worst-case running time of UnionFind with path compression is linear.



2. (4 marks) What is the probability for the 3rd insertion to have exactly two collisions while using linear probing in the hash table?



3. (4 marks) Run EQUALITYREASONING algorithm using UNIONFIND on the following constraints.

$$x_1 = x_2 \wedge x_3 = x_4 \wedge x_5 = x_6 \wedge x_3 = x_6 \wedge x_5 = x_6 \wedge x_1 = x_7 \wedge x_2 = x_4 \wedge x_4 \neq x_2$$

Show state of the disjoint sets after processing each equality. Process them in the order given in the question.



✓ 4. (6 marks) Let us modify the definition of Red-black tree.

Definition 1.1 *A red-black tree T is a binary search tree such that the following holds.*

- *All internal nodes are colored either red or black*
- *Null leaves have no color*
- *Root is colored black*
- **If the parent of a red node is red, then its grand parent cannot be red.**
- *All paths from the root to null leaves have the same number of black nodes.*

Bold text is the modified part of the definition. The black-height is defined as usual.

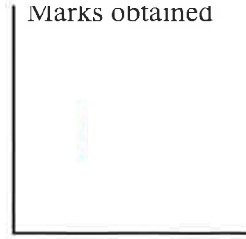
Give the maximum and minimum number of nodes in the above modified red-black tree with black-height h .

M29-LA001-YELLOW-24B0973

Space for only question 4.

Marks obtained





Section B

5. (8 marks) The binary tree representation of general ordered trees is defined as follows. The tree containing only the root node is represented by the empty binary tree. For every non-root node i in the tree, there is a corresponding node $b(i)$ in the binary tree. The left child of $b(i)$ is the node $b(j)$ where j is the first child of i , if it exists, otherwise $b(i)$ has no left child. The right child of $b(i)$ is the node $b(k)$, where k is the right sibling of i , that is the next node having the same parent as i .

Write an efficient algorithm to compute the binary tree representation of a given tree and vice versa. Assume any suitable implementation of trees and binary trees.

M29-LA001-YELLOW-24B0973

Space for only question 5.

Marks obtained





6. (8 marks) Implement doubly linked lists using only one pointer value $x.np$ per item instead of the usual two (next and prev). Be sure to describe what information you need to access the head of the list. Show how to implement the SEARCH, INSERT, and DELETE operations on such a list. (Hint: Exploit the power of bitwise operations)

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Space for only question 6.

Marks obtained





7. (12 marks) Let the sequence $a_1, \dots, a_{n+m}$ be a permutation of $1, \dots, n+m$ such that for each $p \leq q \leq n$, if $a_i = p$ and $a_j = q$ for some i and j , then $i \leq j$. Give the exact recurrence relation $T(n, m)$ for the average insertion time when inserting the sequence into an empty BST. The time is measured by the number of key comparisons performed during the insertions. Show that $T(n, m) \in \Omega(n^2)$.

M29-LA001-YELLOW-24B0973

CHECK YOUR ROLL NUMBER ABOVE.

DO NOT FLIP ANSWERBOOK UNTIL THE EXAM STARTS.
Space for only question 7.

Marks obtained

